

# Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: SAMSUM | Context: South Asia and MENA — Clinical NLP Researchers and AI Practitioners

**Output Ontology definition:** Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Each dialogue receives exactly one reference summary **[Q24]** under a four-point style guide (short, information-extractive, interlocutor-naming, third-person) **[Q23]**. There are no clinical sub-categories — no symptom identification, risk indicators, therapeutic intervention markers, or empathy/alliance categories — and no representation of region-specific counseling outputs (religious reframing, family-mediated resolution plans, stigma-sensitive omissions). Empirical sampling reveals concrete failure modes: suicide discussed as social reassurance trivia **[DATASET-D10]**, abuse-victim self-derogation reproduced as factual summary content **[DATASET-D6]**, peer distress reduced to social transaction **[DATASET-D33]**. The single-reference design forecloses representation of regional practitioner norm divergence. Given HIGH priority weight on OO and user confirmation that region-specific salience matters, this is a severe structural validity violation.

| ID  | Evidence                                                                                                                                                                                                                                                                                                       |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q23 | "After collecting all of the conversations, we asked language experts to annotate them with summaries, assuming that they should (1) be rather short, (2) extract important pieces of information, (3) include names of interlocutors, (4) be written in the third person." (p.2)                              |
| Q24 | "Each dialogue contains only one reference summary." (p.2)                                                                                                                                                                                                                                                     |
| D6  | Pearl's boyfriend mistreats her but she loves him anyway. Patrick is trying to convince her to leave him. Pearl is fat and ugly, so it's not so easy for her. — Summary reproduces a derogatory self-description as factual without clinical reframing — illustrates embedded Western annotator salience norms |
| D10 | Mickey tries to calm her with the fact that more people commit suicide on the Golden Gate Bridge than in that forest. — Suicide statistics treated as social reassurance in summary — no clinical risk annotation or flagging                                                                                  |
| D33 | Francine and Jessie talked yesterday. The talk has been helpful for Jessie. Francine says she's always there for Jessie. Jessie is very thankful. — Summary of peer emotional support — captures who did what but contains no clinical salience categories (distress type, risk level, intervention)           |

## Strengths

- The benchmark's style guide **[Q23]** does enforce interlocutor-naming, which transfers to clinical summarization where speaker attribution of symptoms and statements is clinically important **[DATASET-D20]**.

| ID  | Evidence                                                                                                                                                                                                                                                                          |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q23 | "After collecting all of the conversations, we asked language experts to annotate them with summaries, assuming that they should (1) be rather short, (2) extract important pieces of information, (3) include names of interlocutors, (4) be written in the third person." (p.2) |
| D20 | Jimmy is going to take medication for a month to cure his acute gastritis. — Summary extracts action item and medical fact; third-person, attribution-focused                                                                                                                     |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                   |
|------|-------------------------------------------------------|
| OO-1 | Review output label categories for regional relevance |
| OO-2 | Identify missing regionally specific categories       |

|      |                                                              |
|------|--------------------------------------------------------------|
| OO-3 | Flag categories encoding non-regional values                 |
| OO-4 | Consider stakeholder-driven taxonomy redesign                |
| OO-5 | Document taxonomy issues harming structural/content validity |

**Checklist responses:**

- OO-1:** Output label categories are limited to a single free-form summary per dialogue with implicit salience rules (action items, attribution, factual outcomes). No clinical or regionally relevant categories exist.
- OO-2:** Missing clinical categories include: symptom disclosure markers, risk/safety flags, therapeutic intervention type, alliance markers, family-mediated resolution annotations, religious reframing markers, and stigma-sensitive framing tags. Empirical evidence: suicide content received no risk annotation [DATASET-D10], distress disclosure received no clinical marker [DATASET-D33].
- OO-3:** Embedded values include: action/outcome salience over emotional content, individual rather than family-mediated framing, secular rather than religious framing, direct disclosure rather than face-saving omission. These conflict with documented MENA fatalism/tawakkul framing and South Asian izzat-mediated indirect communication norms.
- OO-4:** Stakeholder-driven taxonomy redesign would be required for valid SA-MENA clinical deployment. WEB-15 (MentalCLOUDS) provides one English-language template with three clinical components (symptom/history, patient discovery, reflection) but is itself American-demographic-limited (WEB-18).
- OO-5:** Structural validity is severely harmed: the construct's structure (clinical summarization) is misrepresented as social transaction logging. Content validity is severely harmed: clinical and regional categories are entirely absent. External validity is harmed: scoring against this ontology will not generalize to clinical practitioner judgments.

| ID     | Evidence                                                                                                                                                                                                                                                                                             |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q23    | "After collecting all of the conversations, we asked language experts to annotate them with summaries, assuming that they should (1) be rather short, (2) extract important pieces of information, (3) include names of interlocutors, (4) be written in the third person." (p.2)                    |
| Q24    | "Each dialogue contains only one reference summary." (p.2)                                                                                                                                                                                                                                           |
| D10    | Mickey tries to calm her with the fact that more people commit suicide on the Golden Gate Bridge than in that forest. — Suicide statistics treated as social reassurance in summary — no clinical risk annotation or flagging                                                                        |
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| WEB-13 | AraHealthQA 2025 found BERTScore 'failed to fully measure appropriateness or trustworthiness' for Arabic mental health QA — direct corroboration that automated metrics are insufficient (https://arxiv.org/html/2508.20047v2)                                                                       |
| WEB-15 | MentalCLOUDS supplements ROUGE/BERTScore with expert evaluation across six clinical parameters — shows the form of evaluation SAMSum lacks (https://mental.jmir.org/2024/1/e57306)                                                                                                                   |
| WEB-18 | MentalCLOUDS itself notes American demographic limitation — even closest English clinical analogue carries cultural caveats (https://arxiv.org/pdf/2402.19052)                                                                                                                                       |

**Information Gaps**

- Whether regional practitioners would converge on a single replacement taxonomy or require sub-regional variants

**Requires Expert Verification**

- Specific clinical and regional categories that SA-MENA practitioners would prioritize in a redesigned ontology

**Remediation**

**Gap 1: Single-reference summary design with messenger-chitchat salience norms; no clinical or regional categories.**

**Recommendation:** Adopt or adapt MentalCLOUDS' three-component schema (symptom/history, patient discovery, reflection) per WEB-15 as a baseline, then add region-specific categories (religious reframing markers, family-mediated resolution, stigma-sensitive omission flags). Move from single-reference to multi-reference design to capture legitimate practitioner variation.

| ID     | Evidence                                                                                                                                                                                                                                 |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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